



4.1 Sampling techniques

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A sample is used to draw conclusions about a population when surveying the whole population is impractical. Simple random sampling gives every member an equal chance of selection; systematic sampling selects every k th member after a random start; stratified sampling divides the population into subgroups (strata) and samples proportionally from each, ensuring subgroup representation; convenience and quota sampling are quicker but more prone to bias. The best method depends on the population structure and practical constraints.

Worked example: follow the reasoning

A school has 300 students: 120 in Year 12 and 180 in Year 13. A stratified sample of size 50 is required. Find how many students should be sampled from each year.

Solution: Year 12: $50 \times (120/300) = 20$ students. Year 13: $50 \times (180/300) = 30$ students.

Quick check · multiple choice

Which sampling method involves selecting every k th member of a population after a random starting point?

- A Stratified
- B Systematic
- C Convenience
- D Quota

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A company has 4 departments with 40, 60, 80 and 20 employees respectively (200 in total). A stratified sample of 20 employees is required for a survey. (a) Calculate the number of employees to be sampled from each department. (b) Explain one advantage of stratified sampling over simple random sampling in this context. (c) State one limitation of stratified sampling. [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Department sizes 40,60,80,20 give samples of $20 \times 40 / 200 = 4$, $20 \times 60 / 200 = 6$, $20 \times 80 / 200 = 8$, and $20 \times 20 / 200 = 2$ respectively. (b) It guarantees each department, including the smallest (20 employees), is proportionally represented, which a simple random sample might fail to achieve by chance. (c) It requires accurate knowledge of each stratum's size in advance and can be more complex to organize.

Quick check answer

B. Systematic sampling selects members at fixed intervals (every k th) from a randomly chosen starting point.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.